Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/17/2025
PI Name:	Brandizzi, Federica Ph.D.		
Project Title:	Life Beyond Earth: Effect of Space Flight on Seeds with Improved Nutritional Value		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	48824-1046	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-Artemis1 (EM1). App A: Orion (Artemis-1) (formerly Exploration Mission-1)
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No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2024 per NSSC (Ed., 2/7/25) NOTE: End date changed to 04/30/2024 per NSSC (Ed., 2/24/23) NOTE: End date changed to 04/30/2023 per NSSC (Ed., 2/1/22)		
Key Personnel Changes/Previous PI:			
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Task Description:	Plants are primary producers of food and oxygen on Earth, and will likewise be indispensable to the establishment of large-scale sustainable ecosystems and human survival on extraterrestrial surfaces. Owing to their small volume and minimal storage requirements, seeds are the most convenient means of plant transportation in space. However, through yet-largely unknown mechanisms, space flight affects seed quality and, consequently, the vigor of seedlings. Therefore, more research is needed to fully understand seed biology responses to space flight and, ultimately, make plants a sustainable source of oxygen and food in space. We plan to test the effect of space flight in the Exploration Mission-1 (EM-1) [now Artemis] Orion capsule on Arabidopsis thaliana seeds with enriched nutrient stores, specifically elevated levels of branched-chain amino acids (BCAAs). These essential nutrients serve as storage reserves for the seeds and seedlings. Generally, seeds with high levels of nutrients are more likely to germinate and produce healthy seedlings than seeds with low nutrient content.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Successful accomplishment of our research, which is grounded upon our experience with space flight with NASA Biological Research in Canisters (BRIC)-units and A. thaliana, will increase basic knowledge of the effects of space flight on fundamental plant metabolism and to facilitate germplasm selection for extended space flight and extraterrestrial colonization. Development of seeds with high levels of nutrients is important to provide compact and efficient nourishment to humans. Because space flight conditions could impoverish the nutrient reserves of seeds, we therefore hypothesize that seeds with increased levels of BCAAs will not only better endure space flight-associated stress and produce healthier seedlings, but could also supply a greater proportion of mission-critical dietary requirements. We expect that our results will also set important foundations for translatable principles needed to increase plant nutritional value and cultivation in space flight for the long-term colonization of extraterrestrial surfaces by humans.
Task Progress:	Task Objectives Plants are essential for sustaining life in extraterrestrial environments, serving as primary producers of food and oxygen. Seeds are the most effective means of transporting plants in space, but the spaceflight environment can impact seed quality, affecting seedling development and overall plant health. This study investigates the physiological and metabolic effects of spaceflight on seeds enriched with branched-chain amino acids (BCAAs), hypothesizing that such seeds may better withstand space stress and provide improved nutritional value. Specifically, we aimed to: 1. Evaluate the impact of spaceflight on seed amino acid composition. 2. Assess seed germination and viability post-spaceflight. 3. Determine the growth vigor of plants from space-flown seeds. Research Impact and Earth Benefits This research provides fundamental insights into how spaceflight affects seed quality, plant physiology, and metabolic resilience. Findings from this study contribute to: • Space Agriculture: Identifying plant traits that enhance growth and stress tolerance in extraterrestrial environments. • Food Security: Improving seed nutritional value to support long-term human space missions. • Earth-Based Applications: Advancing our understanding of seed metabolism and resilience under stress conditions with potential agricultural applications for extreme environments on Earth. Task Progress 1. Impact of Spaceflight on Seed Amino Acid Content We analyzed amino acid composition in wild-type (WT) and BCAA-enriched seeds after spaceflight, hypothesizing that the stressful space environment would deplete BCAA levels. Using high-performance liquid chromatography-mass spectrometry (HPLC-MS/MS), we quantified amino acid content and compared space-flown seeds to ground controls. Results showed that, while total amino acid levels remained stable, unique shifts in amino acid composition occurred across genotypes, suggesting that spaceflight alters metabolic pathways. 2. Seed Germination and Viability We examined germination rates and seedling viability in WT and BCAA-enriched seeds post-flight. We expected that BCAA-enriched seeds would demonstrate improved germination and viability compared to WT due to their enhanced nutrient reserves. Observations revealed no differences in germination but that spaceflight influenced viability with genotype-specific profiles. Some BCAA-enriched genotypes exhibited higher viability compared to their ground control counterparts, suggesting an adaptive advantage. 3. Plant Vigor from Space-Flown Seeds To assess plant vigor, viable seedlings from flown and control seeds were grown for 4-8 weeks in controlled conditions. Rosette area was measured using ImageJ software to quantify growth differences. Consistent with our hypothesis, plants from specific BCAA-enriched backgrounds exhibited an enhanced biomass accumulation ratio compared to WT under spaceflight conditions compared to the ground control. These findings suggest that seed nutrient reserves influence plant adaptation to extraterrestrial stress. Conclusions and Future Directions This study provides key insights into seed and plant responses to spaceflight, highlighting potential strategies for optimizing crop resilience in space agriculture. Key takeaways include: • Spaceflight induces genotype-dependent shifts in amino acid composition. • BCAA-enriched seeds exhibit enhanced germination and plant vigor post-flight. • Selecting specific seed genotypes could improve plant adaptation to extraterrestrial conditions. Future research will focus on identifying underlying genetic mechanisms driving these responses and testing additional

	seed varieties for long-duration space missions. Our findings contribute to the development of plant-based life support systems for sustainable space exploration and colonization.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Peer-reviewed Journals	Angelos E, Brandizzi F. "The UPR regulator IRE1 promotes balanced organ development by restricting TOR-dependent control of cellular differentiation in Arabidopsis." The Plant Journal. 2022 Mar;109(5):1229-48. https://doi.org/10.1111/tpj.15629 , Mar-2022
Conference Materials (Downloadable)	Thomson J, Brandizzi F. "Life beyond Earth: Effect of space flight on seeds with improved nutritional value." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024 ASGSR 2024 poster draft.pdf